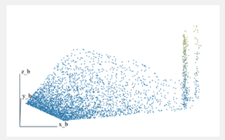
Point-cloud-conditioned UAV trajectory policy with DPPO post-training

Observation Space

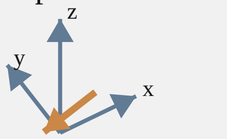


Local XYZ point cloud

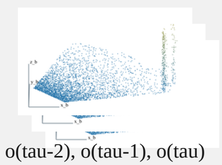
g_b

d = 28.53 m

Relative goal and distance



p, sin(yaw), cos(yaw)



o(tau-2), o(tau-1), o(tau)

Position and heading

Three-frame observations



Environment

Local Navigation

Observe - plan - execute

Conditional Diffusion Policy

9-D state history

State MLP 64-D

XYZ point cloud history

PointNet 128-D

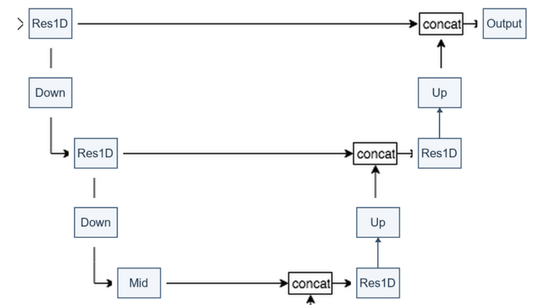
Feature concatenation + time order c(t): 576-D

Noisy trajectory Z(k)

Diffusion step k

Time embedding

Conditional 1D U-Net



FiLM: concat[c(t), emb(k)]

x0 prediction -> DDIM transition -> repeat

DPPO: Environment MDP + Denoising MDP

Environment step t

c(t) 3-frame

Condition

Z(t, K) N(0,I)

...

Z(t, j)

Gaussian transitions

Z(t, 0)

Execute H4

r(t)

Inner process: iterative trajectory denoising

Environment step t+1

c(t+1) 3-frame

Condition

Z(t+1, K) N(0,I)

...

Z(t+1, j)

Gaussian transitions

Z(t+1, 0)

Execute H4

Environment reward Critic / GAE / discount

Denoising transition log-probabilities

Clipped PPO surrogate objective Denoising-step ratio

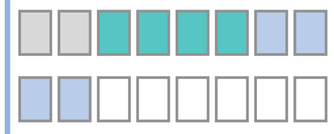
Value loss

Update LoRA

Update critic

Encoder and pretrained backbone remain frozen

Receding-horizon execution



16 predicted elements
2 for alignment
Plan 8 / execute 4

Update observations and replan

U-Net graphic adapted from Yazdani (CC BY-SA 4.0). Inputs/layout: project records. DPPO concept: Ren et al. (2024).